

Last Mile Logistics Auditor

Delivery promise accuracy and customer sentiment audit

DELIVERED ORDERS

analysis base

LATE RATE

late or super late

WORST STATE

23.9% late

Executive Summary

Answering whether delivery failures are regional or nationwide

LATE RATE

8.1%

7,827 late or super late

WORST REGION

AL

23.9% late rate

REVIEW DAMAGE

-2.51

on time vs super late

CATEGORY RISK

Audio

12.9% late rate

Main Finding

- Delivery failures are concentrated in specific states.
- AL, MA, PI, CE, and SE have late rates well above the national average.
- Super-late orders are associated with a steep drop in customer review score.

Business Meaning

- The CEO's over-promising concern is supported by the data.
- Regional promise recalibration should be prioritized.
- Product-category risk gives an extra operational repair path.

Data Model and Cleaning

Schema builder acceptance criteria

Core Joins

- Orders to customers on customer_id.
- Aggregated reviews to orders on order_id.
- Products and translations added for category analysis.

Duplicate Control

- Reviews are grouped to one row per order before joining.
- Row counts and unique order IDs are checked after joins.

Delay Logic

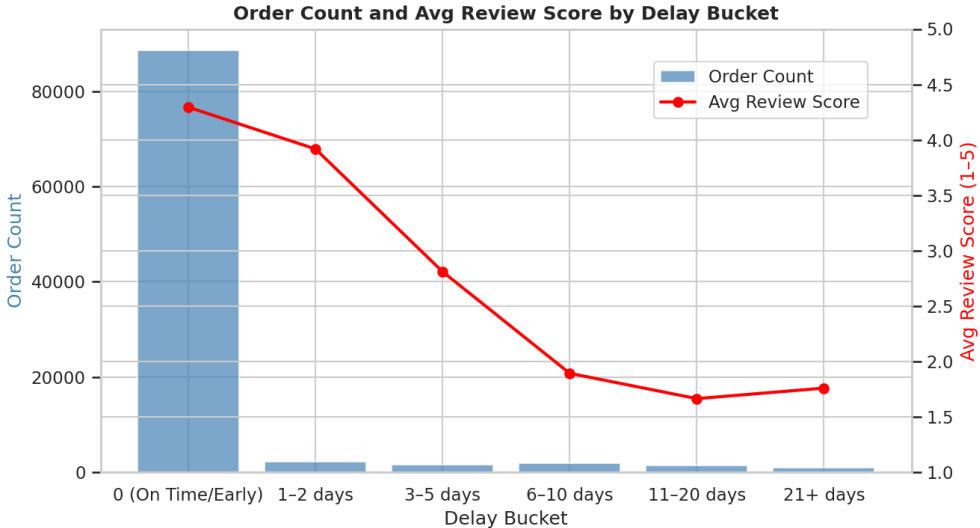
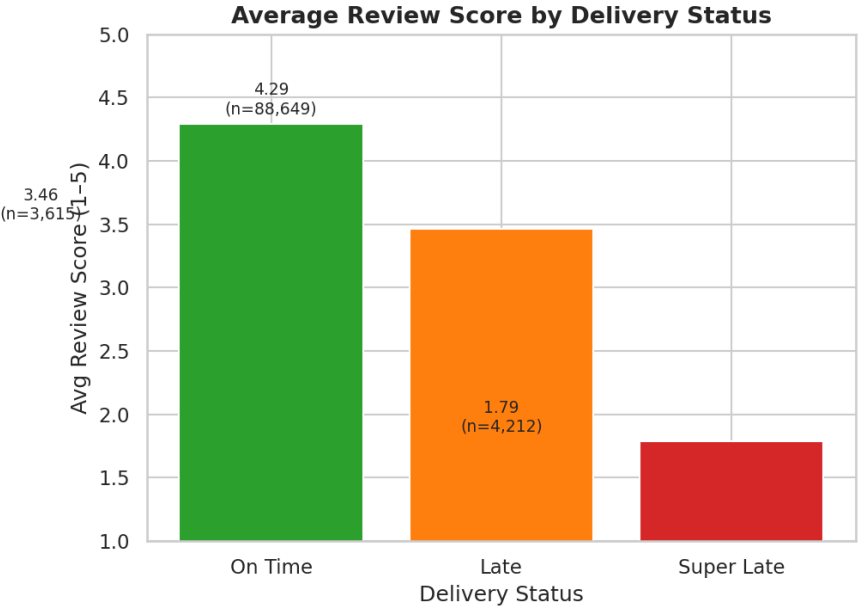
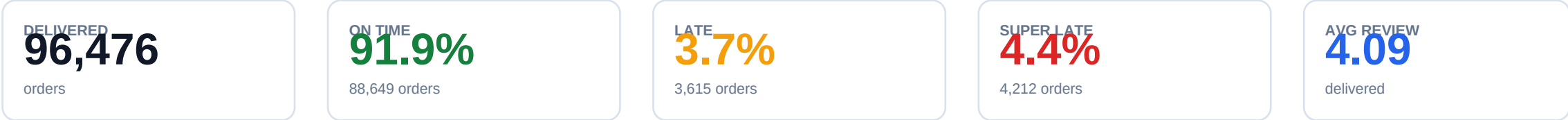
- Days_Difference = estimated date - actual date.
- Undelivered orders are flagged separately.

Delivery Status Rules

- On Time: delivered on or before the estimated date.
- Late: delivered 1 to 5 days after the estimated date.
- Super Late: delivered more than 5 days after the estimated date.

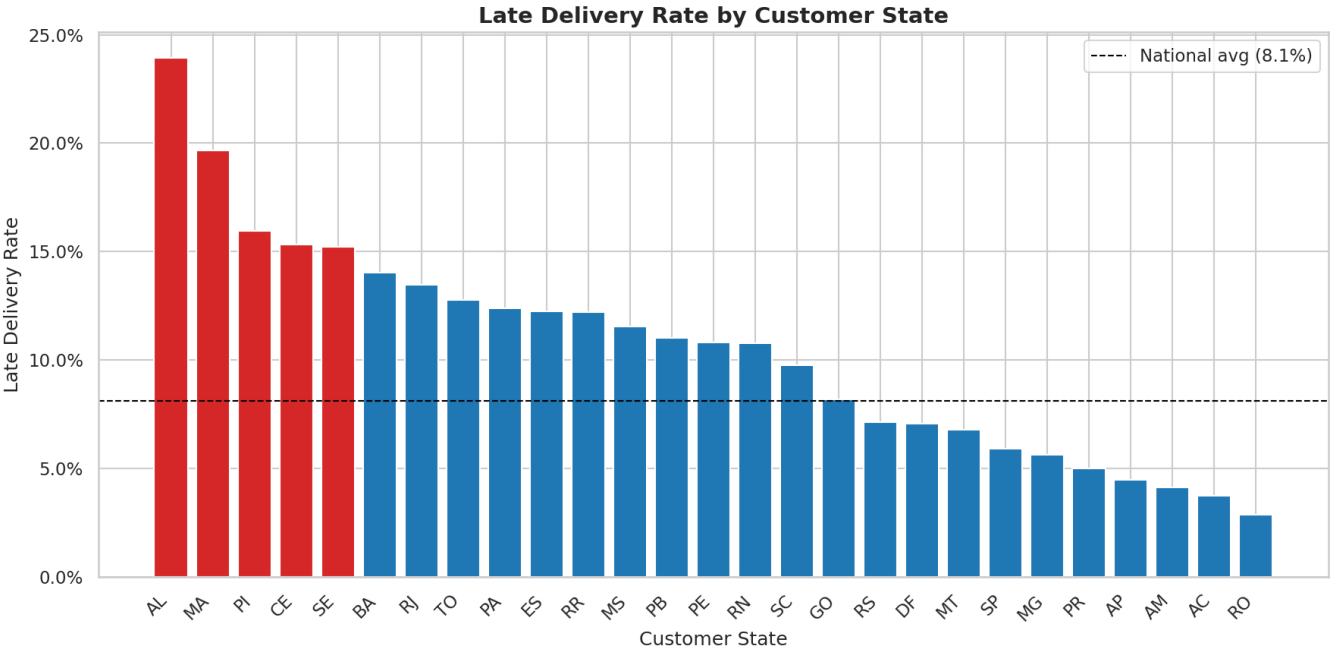
National Delivery Performance

Delivery promise accuracy overview



Regional Delivery Risk

Geographic heatmap story shown as ranked state performance



Highest Late Rates

- AL: 23.9%
- MA: 19.7%
- PI: 16.0%
- CE: 15.3%
- SE: 15.2%

Interpretation

- The problem is concentrated regionally.
- Repair work should start in states above the national benchmark.

Customer Sentiment Impact

Delivery delay vs average review score

ON TIME REVIEW

4.29/5

best experience

LATE REVIEW

3.46/5

moderate damage

SUPER LATE REVIEW

1.79/5

severe damage

GAP

2.51

points

Evidence

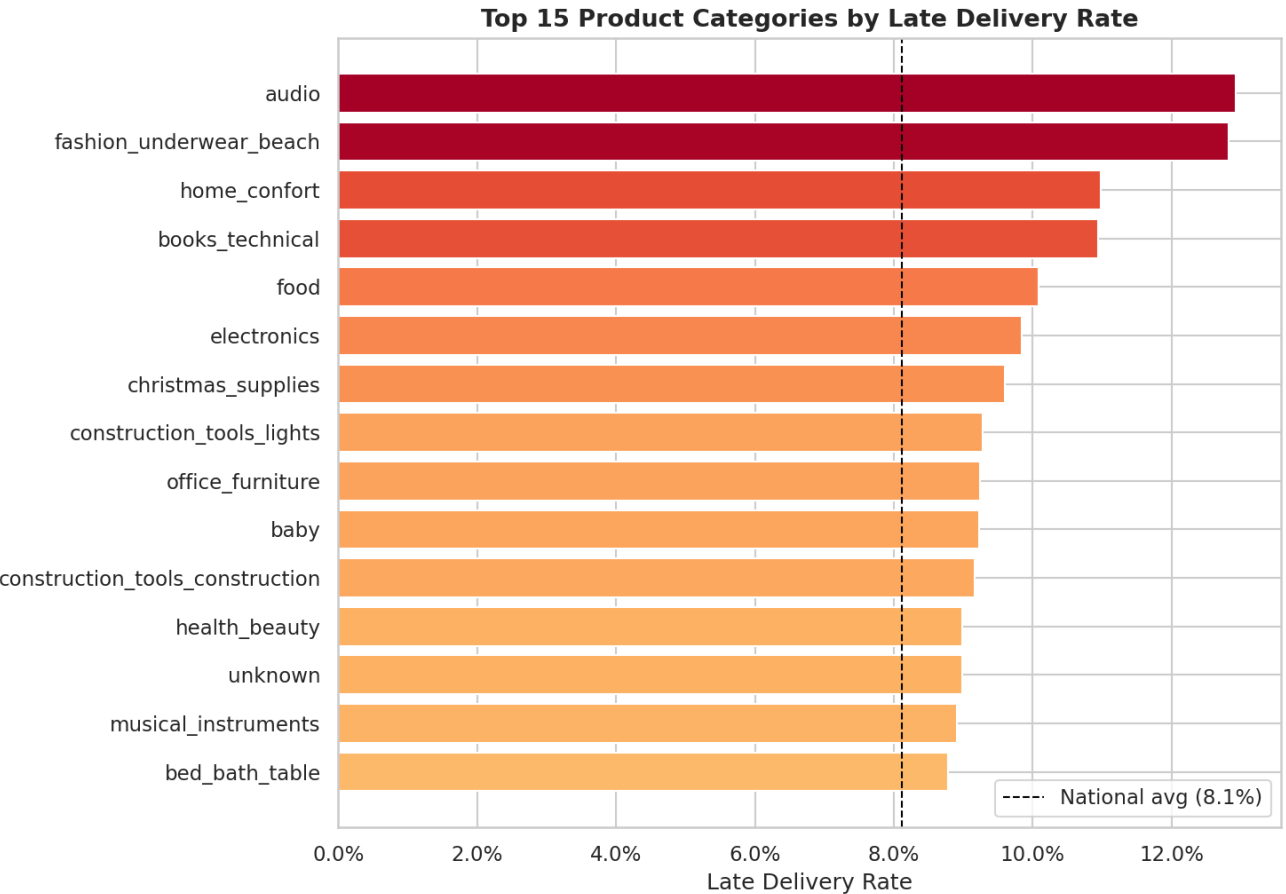
- Average review score falls as delivery performance worsens.
- Super-late orders perform far below on-time orders.
- This supports the CEO's concern about over-promising.

Careful Language

- This is observational analysis, so use associated with rather than caused by.
- The pattern is still strong enough to guide operations.

Candidate's Choice: Product Category Risk

Translated English category analysis



Top Category Risk

- Audio has the highest late rate at 12.9%.
- Fashion underwear beach follows at 12.8%.
- Home comfort follows at 11.0%.

Business Value

- Category analysis reveals product and fulfillment complexity.
- It adds an improvement path beyond geography.

Dashboard Walkthrough

Public Streamlit audit tool

Executive KPIs

- Delivered volume, promise accuracy, late rate, super-late rate

Regional Drilldown

- State selector, regional late-rate ranking, and risk matrix.

Category Drilldown

- Category risk ranking, impact matrix, and leaderboards.

Dashboard Link

- <https://amalitech-pzzepkval4rfh3phyf2jh.streamlit.app/>
- Built from summary CSVs only, so the raw Kaggle dataset is not required for deployment.

Recommendations

Operational actions for Veridi Logistics

1. Prioritize Regional Repair

- Start investigations in AL, MA, PI, CE, and SE.
- Compare carrier, routing, and fulfillment processes in these states.

2. Recalibrate Delivery Promises

- Use more conservative estimates in high-risk states.
- Track promise accuracy as an executive KPI.

3. Track Super-Late Separately

- Super-late orders are associated with the steepest review damage.
- Escalate orders once they pass five days late.

4. Review Product Categories

- Investigate audio and other high-risk categories.
- Check packaging, seller behavior, and carrier handling.

Submission Checklist

Final reviewer-facing deliverables

Included in Repository

- README with executive summary and technical notes.
- Notebook and HTML notebook export.
- Streamlit app and requirements file.
- Dashboard-ready outputs and static chart exports.

Public Links to Test

- GitHub repository.
- Streamlit dashboard.
- Presentation PDF.
- Optional video walkthrough if created.

Final CEO Answer

- Delivery failures are concentrated in specific regions and are strongly associated with lower customer reviews, especially when orders become super late.